

# Han-Teng Liao

Socio-Technical System Architect and Researcher (former Associate Professor and Vice Dean)

Taipei and Penang

✉ Email 🏠 Project 🆔 0000-0003-1081-5599 📄 Google Scholar 🌐 Web of Science 🐙 GitHub 🔗 LinkedIn 🐦 twitter

*Drives global value chain transitions through data-driven innovations, focusing on engineering safe and sustainable systems that span the industrial lifecycles: from the AI and semiconductor nexus, back to upstream petro-chemical inputs, and forward to downstream digital economies.*

## Professional Service

### Peer Reviewers

*Editorial*

Completed 25+ reviews (mostly Q1 and Q2 journals), esp. on AI and Industry 5.0.

*2021 — Present*

- **Scientific Reports** (Q1:2) **Financial Innovation** (Q1:3); **Nature Humanities and Social Sciences Communications** (Q1:1)
- **Politics and Governance** (Q2:1), **Frontiers in Public Health** (Q1:1); **Journal of Clinical Epidemiology** (Q1:1)
- **Heliyon** (Q1:2); **Cogent Economics & Finance** (Q1:1); **Cogent Education** (Q2:1); **Applied Economics Letters** (Q3:1)
- **Future Internet** (Q2:1); **Discover Artificial Intelligence** (1); **Discover Education** (1)
- **Sustainability** (Q2:4); **Environment Systems and Decisions** (1)

### Technical Committee Member

*Technical*

**Joint IEEE Systems Engineering/TEMS TC on Carbon Neutrality**

*2026 — Present*

- Explore active research and strategies to promote carbon neutrality best practices, driving consortium actions with bodies like **SiNANO** and orchestrating strategic liaisons across IEEE Technical Committees and international organizations.

### Expert Contributor, International Telecommunication Union

*ITU*

ITU-T Focus Group on AI for Agriculture (FG-AI4A)

*2023 - 2024*

- Authored technical input document b-FG-AI4A-I-132 on linking carbon accounting data with agricultural business.
- Successfully integrated the framework into ITU-T Y.Suppl.83 (07/2024) under Section 7.2.7 (TG-Digital Agri-Finance).
- Translates ‘Carbon Information Chains’ regional best practices into international standard reference.

### Student Branch Development & Launch

*Guangzhou, China*

IEEE Technology and Engineering Management Society (TEMS)

*2019 - 2023*

- Orchestrated and managed the establishment of the IEEE TEMS Student Branch during tenure as Center Director.

### Program Committee Chair (MSIED)

*Guangzhou, China*

Management Science Informatization and Economic Innovation Development Conference

*December 18-20, 2020*

### Conference Special Session Chair

*Guangzhou, China*

World Eco-Design Conference (WEDC) 2019

*2019*

- Chaired the session on **Internet Eco-Design: Sustainable Development Innovation Design and Internet Eco-systems** (第二届世界生态设计大会-互联网生态设计：可持续发展创新设计与互联网生态).

### Program Committee Member

*Yogyakarta, Indonesia*

14th International Conference of IFIP Working Group 9.4

*2017*

### Invited Expert

*United Nations ESCAP*

Expert Meeting on Big Data and the 2030 Agenda for Sustainable Development

*2015*

- Provided expert insights on leveraging big data to advance the UN 2030 Sustainable Development Goals (SDGs).

### Visiting Expert

*Leiden University, Netherlands*

ERC Project “Authoritarianism 2.0”

*2014*

- Invited by Professor Daniela Stockmann to advise the European Research Council (ERC) project team on digital methods.

### Conference Organizer

*Washington DC, US*

Chinese Internet Research Conference (CIRC 2011)

*2011*

- Co-organized the conference theme “Chinese Factors in Internet Development: Global Public Goods, National Policies, and Private Interests.”

Education

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|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>DPhil (Ph.D.) in Information, Communication, and Social Sciences</b><br>University of Oxford                                            | <i>Oxford, United Kingdom</i><br>10/2007 - 10/2014 |
| <b>M.Sc. in Computer Science and Information Engineering</b><br>National Taiwan University                                                 | <i>Taipei, Taiwan</i><br>01/08/2000 - 01/06/2004   |
| <b>M.S.S. (Master of Social Science) in Journalism</b><br>National Taiwan University                                                       | <i>Taipei, Taiwan</i><br>01/08/2000 - 01/06/2004   |
| <b>B.Sc. in Electrical Engineering</b><br>National Taiwan University (Includes a year-long exchange at the University of British Columbia) | <i>Taipei, Taiwan</i><br>01/10/1994 - 01/06/2000   |
| <b>Bachelor of Arts in Foreign Languages and Literatures</b><br>National Taiwan University                                                 | <i>Taipei, Taiwan</i><br>01/08/1995 - 01/06/2000   |

Work Experience

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| <b>Editor &amp; Project Founder (Public Service Initiative)</b><br>scvcs.footprints.cv - SSbD Chips Value Chain Services   GBS.today                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <i>Penang, Malaysia</i><br>01/01/2026 - Current |
| <b>Data Scientist and CEO</b><br>Oxford Roadmapping <ul style="list-style-type: none"><li><b>Ecosystem Innovation Visualization:</b> Designed and deployed specialized visualization prototypes—including <b>visNetZero</b> (decarbonization pathways), <b>visCEADs</b> (carbon emission datasets), and <b>NetBib</b> (bibliometric networks)—to align industrial capabilities with SDG targets and operationalize ITU Digital Innovation Toolkits.</li><li><b>Co-Principal 2022ZDJS121 “Roadmapping for Precision-Decarbonization”:</b> Served as Co-Principal Researcher, Lead Data Scientist, and Corresponding Author for the "Roadmapping for Precision-Decarbonization" project, integrating qualitative policy analysis with quantitative precision analytics.</li><li><b>Data Science for Strategic Roadmapping:</b> Orchestrate workflows transforming raw scientometric, bibliometric, and patent data into strategic narratives to guide regional industrial policies, such as the EU Safe and Sustainable by Design (SSbD) framework and ASEAN Digital Economy Framework Agreement (DEFA).</li><li><b>High-Impact Bibliometric Publications:</b> Authored 4 SCIE/ESCI-indexed (Q2) peer-reviewed articles de-risking green digital infrastructure investments, establishing frameworks for: <b>Digital Transformation &amp; Business Ecosystems, Partner Maps for ESG Capability Building, ITU-Standard Smart Digital Platforms for Carbon Neutral Management, and Supply Chain Resilience &amp; Human Rights Knowledge Mapping.</b></li><li><b>Full-Stack De-toxification &amp; SSbD:</b> Applied the Safe and Sustainable by Design (SSbD) framework</li></ul> | <i>Guangzhou, China</i><br>01/01/2023 - 2025    |
| <b>Director, Higher Education Impact Assessment Center</b><br>Nanfang College of Sun Yat-Sen University <ul style="list-style-type: none"><li><b>Strategic Institutional Growth:</b> Founded and led a university-wide research center to optimize institutional impact and rankings (QS/THE), resulting in the center producing the highest volume of indexed research output in the institution.</li><li><b>Data-Driven Quality Assurance:</b> Architected performance visualization systems to monitor Outcome-Based Education (OBE) metrics, advocating international standards such as the Washington Accord (IEET).</li><li><b>Interdisciplinary AI Integration:</b> Orchestrated cross-departmental research projects (Engineering, Management, Accounting, and Humanities) focused on AI-SDG EcoInnovations and green digital transformation.</li><li><b>Research Orchestration:</b> Managed a research portfolio (e.g., 2019GXJK186) exceeding 200,000+ RMB, initiating multi-disciplinary reporting for UN Academic Impact, alongside the launch and development of the IEEE TEMS Student Branch.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <i>Guangzhou, China</i><br>12/2019 — 01/2023    |
| <b>Associate Professor &amp; Vice-Dean, Internet and New Media</b><br>Nanfang College of Sun Yat-Sen University <ul style="list-style-type: none"><li><b>AI Product Management &amp; Curriculum:</b> Acted as Lead Product Manager for a "Venture-Ready" GitHub portfolio, overseeing the end-to-end lifecycle of 18 core courses designed to scale capacity for AI solution leaders and product managers.</li><li><b>Award-Winning “Smart App Design Innovation Research in the Age of New Business, Arts and Engineering Disciplines [2019GXJK186]”:</b> Architected province-level "First-Class" Capstone courses (APIs, Machine Learning) using GitHub-based workflows and automation to integrate systems thinking and design thinking for the Internet industry.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <i>Guangzhou, China</i><br>01/2017 — 01/2021    |

- **Quality Assurance:** Led the transition to Outcome-Based Education (OBE) aligned with IEET/Washington Accord standards, using AI Product Manager competencies to align with international education standards.
- **Curriculum-as-MVP:** Designed a modular degree program where students developed PRD, MRD, and BRD documentation and functional prototypes, enabling graduates to transition into UX Design, Front-end Dev, and AI leadership roles.
- **Agile Leadership & Pedagogy:** Recruited and mentored a 7-person faculty team using Agile and Scrum methodologies to iterate on educational delivery, resulting in a reflexive pedagogical framework that allowed high-performing students to accelerate their graduation timeline by one full year (three students). One student was admitted to and graduated from the **European Institute of Innovation and Technology (EIT)**.

## Research Fellow

United Nations University (UNU) Institute on Computing and Society

Macau, China

01/07/2015 - 01/11/2016

## Yahoo! Fellow, Institute for the Study of Diplomacy

Georgetown University

Washington DC, United States

01/08/2010 - 01/07/2011

## Doctoral Research Fellow

Institute of Sociology, Academia Sinica

Taipei, Taiwan

01/08/2009 - 01/06/2010

## Political Warfare Officer (Mandatory Service)

Taiwan Military

Taiwan

01/11/2005 - 01/12/2006

## Project Manager

Open Source Software Foundry (OSSF), Institute of Information Science, Academia Sinica

Taipei, Taiwan

01/10/2004 - 30/11/2005

## Action Researcher & Graduate Teaching Assistant

Institute of Journalism, National Taiwan University

Taipei, Taiwan

01/09/2002 - 01/12/2003

## Programmer & Independent Developer (Digital Publishing)

Yuan-Liou Publishing Co., Ltd.

Taipei, Taiwan

01/07/2000 - 31/08/2000

## Awards

### Certificate of Merit

Recognized for outstanding academic contributions

IEEE TEMSMET Conference

2021

- Awarded for CSR (Corporate Social Responsibility) and ESG (Environmental, Social, and Governance) reporting research.

### Province-Level 'First Class' Course, Guangdong Department of Education

Curriculum Design & Pedagogy Excellence

Guangdong

2020

- Recognized for the development and instruction of the Capstone Project: 'API, machine learning and AI'.

### Nation-Level First Prize, National College Art & Design Awards

Instructional Excellence

China

2020

- Awarded for Best Teaching for Capstone Project: 'API, machine learning and AI'.

### OxTALENT Awards

Best Interactive Infographics

University of Oxford

2014

- Recognized excellence in university teaching using interactive digital content.

### Best Student Paper Award, Chinese Internet Research Conference

Research Excellence

University of Pennsylvania

2009

- Awarded for foundational digital community research.

### PGP Corporation and Oxford Internet Institute (OII) Scholarship

Merit-based Doctoral Scholarship

University of Oxford

2007

- Value: £12,000 to support DPhil research.

### Taiwan Merit Scholarship

Doctoral Research Fellowship

Taiwan Government

2007

- Awarded for doctoral research in Digital Content (TWD \$2,000,000).

## Publications

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### **From Stacks to Circuits: A Regenerative Socio-Technical Roadmap for AI Infrastructure within Planetary Boundaries**

*ICE IEEE/ITMC, Porto - Portugal*

Conference Paper

*22-25 June 2026*

- Contributors: **Han-Teng Liao**; Karen Ang
- Challenges Nvidia's "electrons-to-tokens" 5-layer scaling trajectories by converting thermodynamic entropy and Scope 3 metrics into dynamic, regeneratively accountable circuits.
- Introduces a Regenerative Socio-Technical (RST) Roadmap to address the hidden environmental, thermal, cognitive and material costs of unconstrained Generative AI scaling.
- (To be) Presented in **Digital Circular Economy** Special Session, providing the conceptual foundation for the **SCVCS** to anchor its **AuditSoS** Pillar through metrology-grounded Digital Product Passports (DPP).

### **Scoping Review of AI, Metrology, and ESG in the Semiconductor Sector: Implications for Safe and Sustainable by Design (SSbD)**

*ICE IEEE/ITMC, Porto - Portugal*

Conference Paper

*22-25 June 2026*

- Contributors: Karen Ang; **Han-Teng Liao**
- Proposes a novel 6-layer System of Systems (SoS) architecture to integrate AI-driven metrology, supply chain ESG metrics, and federated industrial data spaces into an integrated architecture.
- Bridges the gap between precision manufacturing and verifiable environmental sustainability by embedding SSbD as core operational logic for Industry 5.0.
- (To be) Presented in **Technology** Session, providing the conceptual foundation for the **SCVCS** to integrate RealSoS, GuardSoS and AuditSoS Pillars as a System of Systems (SoS).

### **Orchestrating the Twin Transition in Multinational Corporations: Technology Roadmapping for Green and Digital Global Business Services**

*ICE IEEE/ITMC, Porto - Portugal*

Conference Paper

*22-25 June 2026*

- Contributors: **Han-Teng Liao**; Karen Ang
- Synthesizes Technology Roadmapping (TRM) and the ITU innovation toolkit to treat Global Business Services (GBS) units as strategic "operational airlocks."
- Establishes a socio-technical framework for multinational corporations to harmonize internal digital agility and AI-native transitions with external sustainability mandates.
- (To be) Presented in **Innovation** Session, providing the conceptual foundation for the **SCVCS** to cross-map carbon border adjustment data aggregation under its **AuditSoS** Pillar for regional GBS 5.0 networks.

### **Collaborating on ESG consulting, reporting, and communicating education: Using partner maps for capability building design**

*Frontiers in Environmental Science*

Journal article

*March 2023*

- Contributors: **Han-Teng Liao**; Chung-Lien Pan; Yuan Zhang
- Applied design science and circular economy frameworks to map the specific data and digital skills required for ESG consulting and reporting.
- Establishes a collaborative roadmap for universities and industry to build capability for sustainable innovation across disciplines and sectors.

### **Knowledge Mapping of Resilience and Human Rights in Supply Chains**

*Frontiers in Environmental Science*

Journal article

*April 2023*

- Contributors: Chung-Lien Pan; **Han-Teng Liao** (Corresponding Author); Yuan Zhang
- Developed a roadmapping taxonomy to integrate human rights and resilience into supply chain architecture.
- Provides a governance framework for the Twin Transition, offering strategic tools for risk mitigation and value-based alignment in complex, cross-border industrial ecosystems.

### **Knowledge Mapping Analysis of Intelligent Ports: Research Facing Global Value Chain Challenges**

*Systems*

Journal article

*2023*

- Contributors: **Han-Teng Liao**; Tsung-Ming Lo; Chung-Lien Pan
- Utilized scientometrics and science mapping to analyze the digital transformation of intelligent ports within Global Value Chains (GVC).

- Identifies technical and governance bottlenecks in open and collaborative innovation, providing a strategic roadmap for port-city digital integration.

## Carbon neutrality policies and technologies: A scientometric analysis of social science disciplines'

*Frontiers in Environmental Science*

Journal article

October 2021

- Contributors: Yuan Zhang; Chung-Lien Pan; **Han-Teng Liao**
- Conducted a comprehensive science mapping and systematic review of 907 social science articles to delineate disciplinary boundaries and collaborative frameworks for net-zero pathways. - Highlighted the critical intersection where green digital transformation and datafied environmental indicators must guide human production and consumption patterns.

*Proceedings of the 6th International Conference on Humanities and Social Science Research (ICHSSR 2020)*

## A Literature Review of Museum and Heritage on Digitization, Digitalization, and Digital Transformation'

Conference paper

May 2020

- Contributors: **Han-Teng Liao**; Man Zhao; Si-Pan Sun
- Executed a scientometric analysis of 1,107 Web of Science articles using VOSviewer and custom Python scripts to categorize distinct layers of socio-technical transitions. - Mapped the shift in cultural institution research fronts from technology-centric concerns to people-centric concerns (digital literacy, organizational adaptation).

*IEEE Technology and Society Magazine*

## Conflict and consensus in the Chinese version of Wikipedia

Journal article

June 2009

- Contributors: **Han-Teng Liao**
- Investigated the socio-technical architectures, editorial policy alignments, and hyperlinked structures used to govern consensus in high-conflict, cross-border digital environments. - Analyzed how political and cultural cybernetics operate when collaborative web spaces cross sensitive jurisdictional and ideological boundaries.

## More publications and citation indices are available via my ORCID Profile and Google Scholar Profile.

*Academic Profiles*

Impact Information

Current

- Includes comprehensive tracking of peer-reviewed articles, conference papers, and h-index metrics.
- Maintained an accelerating citation trajectory with 800+ total citations (630+ since 2021), achieving an h-index of 16 and an i-index of 24 across leading high-impact sustainability and innovation journals.

## Skills

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### Strategic Architecture & Governance

Integraton of the Socio-Technical: IEEE Systems Council (Social Implications of Technology) and Oxford Internet Institute Multi-disciplinary Training

- AI System & Platform Architecture, ICT System Architecture & Analysis, Systemic Design Thinking, Software Specification Analysis, Architecture Generalization, Knowledge Management

### Intelligence & Foresight (Data Science)

Integraton of Scientometrics/Webometrics and Technology Roadmapping (TRM)

- Strategic Foresight, Scenario Planning, Scientometrics, Bibliometrics, Webometrics, Science Mapping (Bibliometrix, VOSviewer)

### AI, Data Architecture & Automation

Integraton of Open Source/Open Content Ecosystems with Open Science Best Practices

- RAG (Retrieval-Augmented Generation), Harness Engineering, Context Engineering, LLM-Driven Strategic Intelligence, Python (Pandas/NumPy), Semantic Data Modeling, Resource Description Framework (RDF), CMS (Quarto & Obsidian)

### Product & Agile Leadership

Integraton of 2017-2023 Undergraduate Program Reform in China for Industry-ready Talents.

- Full-Lifecycle Product Management (MVP to Scale), PRD/MRD/BRD Documentation, UX/UI Design Research, Scrum, Agile Workflows, GitHub-Based Collaborative Engineering

## **Language Skills**

Deep Integraton of Cross-lingual and Cross-cultural Communication Skills, including the Language Industry.

- Chinese (Mother tongue), English (Fluent — C2 across Listening, Reading, Spoken Production, Spoken Interaction, and Writing)